

Zerve ODSC Datathon: Dataset Progress Report

This report summarizes the data understanding work completed so far on the Zerve events dataset. It focuses on schema discovery, missing-value structure, grouped-null interpretation, and implications for leakage-safe funnel and modeling work.

Dataset Snapshot

- Original shape observed: 3,509,628 rows x 83 columns.
- Each row appears to represent one product analytics event tied to a person, timestamp, and event name.
- The schema includes person properties, browser/device metadata, geolocation fields, AI usage metadata, product activity fields, billing/credit fields, marketing attribution, page engagement fields, and web-vitals metrics.
- A dtype warning appeared during CSV load because several property columns contain mixed types. This is expected in event exports, where sparse properties can vary by event type.

Current Interpretation

The dataset should be treated as an event log, not as a flat user table. Many null values are likely structural: a property is missing because it does not apply to that event. Missingness itself can become a useful signal, especially after aggregating events to the person or workspace level.

Schema Understanding

Major Column Families Found

- Core identifiers: person_id, timestamp, event. These have no missing values and define the event grain.
- Person properties: cloud provider, purpose, role, source, and work type. These describe user profile or onboarding context.
- Browser/device/geography: browser, language, OS, device type, screen size, country, continent, and time zone.
- AI interaction properties: input tokens, output tokens, latency, model, provider, tool call count, and tools called.
- Product activity properties: canvas_id, workspace_id, block types, file type/extension, requirements status, tool name, button/link names, and object counts.
- Credit/subscription properties: credits used/received/remaining, total credits, add-on credits, amount, subscription type, and seats.
- Marketing attribution: UTM source, medium, campaign, content, and term.
- Page engagement/performance: previous pageview scroll/content metrics and web-vitals values such as CLS, FCP, INP, and LCP.

Important Data Understanding Point

The same event row will not contain every property. For example, an AI event may contain model/token/latency fields, while a pageview may contain scroll fields, and a billing event may contain credit fields. This makes grouped missingness a useful way to infer subsystems and event-specific feature families.

Missing-Value Analysis

Method Used

A custom null-summary dataframe was created by iterating through `df.columns`, counting nulls per column, and transposing the result so original columns appeared as rows. The analysis then grouped columns by identical missing-value counts.

Key Output: Repeated Missing Counts

Null Count | Missing % | # Cols | Likely Group

3,477,639	99.09%	8	Previous pageview engagement
2,535,841	72.25%	6	Browser/device metadata
2,959,810	84.33%	5	AI usage metadata
0	0.0%	3	Core event identifiers
1,986,064	56.59%	3	Geo enrichment
3,507,488	99.94%	2	Credit balance snapshot
3,296,653	93.93%	2	AI tool usage
3,509,624	100.0%	2	Feature/subscription signals

Grouped Missingness Inferences

Previous pageview engagement

Columns: `properties.$prev_pageview_last_content`, `properties.$prev_pageview_last_content_percentage`, `properties.$prev_pageview_last_scroll`, `properties.$prev_pageview_last_scroll_percentage`, `properties.$prev_pageview_max_content`, `properties.$prev_pageview_max_content_percentage`, `properties.$prev_pageview_max_scroll`, `properties.$prev_pageview_max_scroll_percentage`

Inference: These fields likely come from the same pageview tracking subsystem. They measure prior page scroll/content exposure and are only populated for a small subset of pageview-like events.

Browser/device metadata

Columns: `properties.$browser`, `properties.$browser_language`, `properties.$device_type`, `properties.$screen_height`, `properties.$screen_width`, `properties.$lib_rate_limit_remaining_tokens`

Inference: These fields appear together because they are client-side tracking metadata. They can support segmentation but are weaker than product behavior for activation modeling.

AI usage metadata

Columns: `properties.$ai_input_tokens`, `properties.$ai_latency`, `properties.$ai_model`, `properties.$ai_output_tokens`, `properties.$ai_provider`

Inference: These columns likely identify AI interaction events. Missingness is meaningful: absence probably means the row is not an AI event rather than bad data.

Core event identifiers

Columns: `person_id`, `timestamp`, `event`

Inference: These are complete and should form the base grain of analysis: who performed what event and when.

Grouped Missingness Inferences Continued

Geo enrichment

Columns: `properties.$geoip_continent_name`, `properties.$geoip_country_name`, `properties.$geoip_time_zone`

Inference: These are inferred location fields. They appear together and can be used for regional segmentation, with an Unknown category for missing values.

Credit balance snapshot

Columns: `properties.credits_remaining`, `properties.total_credits`

Inference: These are highly sparse billing/credit fields. They may be powerful but are leakage-sensitive if predicting monetization or credit usage.

AI tool usage

Columns: `properties.$ai_tool_call_count`, `properties.$ai_tools_called`

Inference: These likely represent agent/tool behavior inside AI workflows. Presence can become a useful product-engagement flag.

Feature/subscription signals

Columns: `properties.feature_tag`, `properties.subscription_type`

Inference: These are almost entirely missing and likely tied to rare subscription or feature-gating events. Treat carefully because they may encode downstream outcomes.

Implications for Feature Engineering

What This Means

- Do not blindly drop high-null columns. In an event log, sparse fields often identify meaningful event types.
- Create presence flags for important sparse groups, such as `has_ai_usage`, `has_ai_tool_call`, `has_credit_snapshot`, `has_pageview_engagement`, and `has_subscription_signal`.
- Aggregate row-level events to `person_id` or `workspace_id` before modeling. Useful features include total events, unique event count, active days, first event, last event, AI event count, total tokens, average latency, canvas/file/block activity, and credit-related activity.
- Use grouped missingness to build feature families. Columns with identical null counts likely originate from the same product subsystem or event instrumentation path.
- Handle categorical missing values as Unknown or Not Applicable, depending on whether missingness means unavailable user metadata or not relevant for the event.

Useful Presence-Flag Examples

- `has_ai_usage` = `properties.$ai_model` is not null
- `has_ai_tool_usage` = `properties.$ai_tool_call_count` is not null
- `has_credit_snapshot` = `properties.credits_remaining` is not null
- `has_pageview_scroll_data` = `properties.$prev_pageview_max_scroll` is not null
- `has_subscription_signal` = `properties.subscription_type` is not null

Leakage and Modeling Plan

Leakage Considerations

Several strong-looking fields may be downstream of the thing we eventually want to predict. For example, credit balance, subscription type, total credits, or feature gating fields should not be used to predict monetization if they are only known after monetization happens.

Recommended Prediction Setup

- Use a time-windowed setup: first 24 hours or first N events as features, then predict later activation or engagement.
- Define activation behaviorally, for example: a user creates or uses a canvas, interacts with AI, uses tools/blocks, uploads files, or returns across sessions.
- Keep the model explainable. Logistic regression, random forest, or gradient boosting can work, but feature importance and actionable interpretation matter more than raw score.
- Evaluate with ROC-AUC, precision/recall, F1, and confusion matrix, but connect the results back to product decisions.

Next Analytical Step

For each grouped-null family, check which event values populate the group. This will validate whether the inferred groups map to pageview events, AI events, credit events, or subscription/feature events. That validation directly supports the rubric categories for data understanding, funnel design, transition logic, and leakage handling.